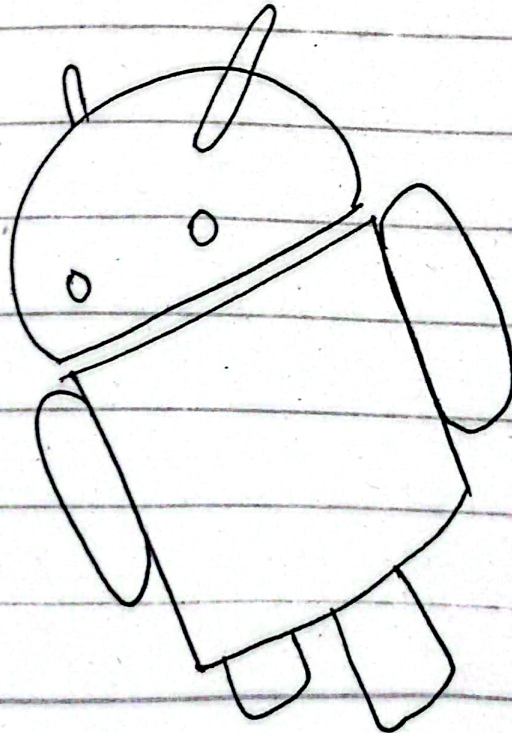


2415

(CSEC-312)
Mobile Application
Development



Ch#1 :- Android Introduction

Q. What is android? ^{↑ - open source - free of cost}
^{↑ - Linux based for mobile devices}

⇒ Android is an open source platform developed by the Open Handset Alliance,

which consists of 3 things: - led by Google & other

(1) OS -

(2) Middleware

(3) Android/Key Applications ^{first beta version of the android SDK was in 2007}

Q. What is difference b/w platform & OS?

⇒ Platform = OS + other stuff

⇒ Android platform = Android OS (Linux kernel) + Java stuff

Q. What is structure or architecture of Android?

⇒ The android architecture has

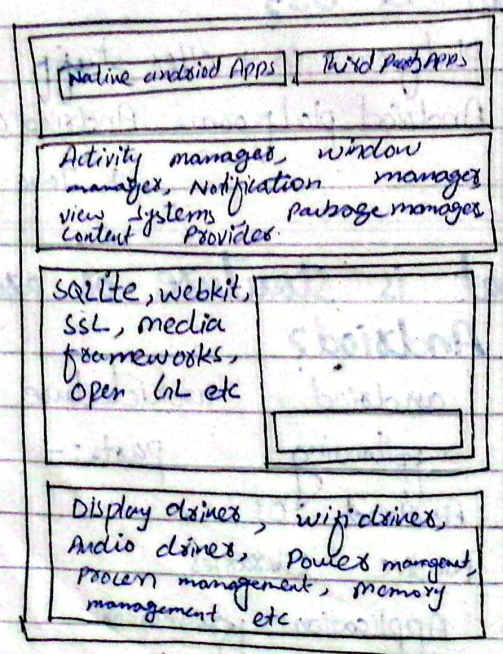
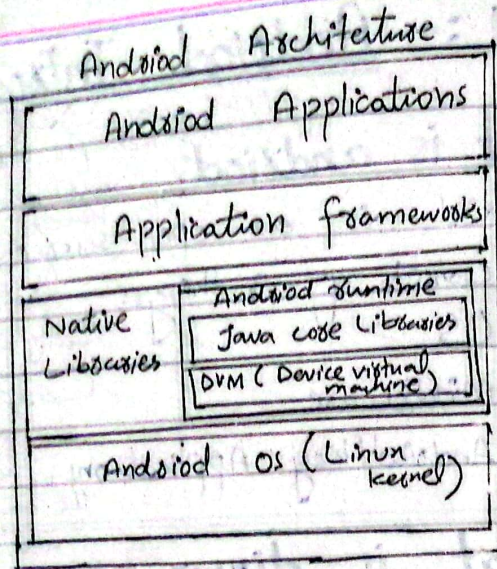
the following parts: -

o Android OS

o Native Libraries

o Application Frameworks

o Android applications



The features of Linux kernel:

- security
- memory management
- Process management
- Network stack
- Drivers model

* Linux kernel:

⇒ At the bottom of the layers is Linux - Linux 3.6 with approximately 115 patches. it is the heart of android architecture.

⇒ This provides a level of abstraction between the device hardware and it contains all the essential hardware drivers. it manages all the available drivers.

* Libraries:

⇒ On the top of linux kernel there is a set of libraries including open-source web browser engine, webkit etc.

* Android Libraries:-

⇒ This category contains those java-based libraries that are specific to android development. And C/C++ core libraries.

Example:

⇒ Application framework libraries, Those that facilitate user interface building; graphics drawing and database access.

⇒ Some key core Android libraries are as follows:-

- | | |
|--------------------|---------------------|
| o android.app | o android.os |
| o android.content | o android.graphics |
| o android.database | o android.provider |
| o android.opengl | o android.telephony |
| o android.view | o android.media |
| o android.widget | o SQLite |
| o android.webkit | o FreeType |
| o android.text | o SSL |
| o android.util | |

* Android Runtime:-

⇒ This is the third section of the architecture and available on the second layer from the bottom.

⇒ This contains a key component called Dalvik virtual machine which is a kind of java virtual machine specially designed and optimized for Android.

⇒ The Dalvik VM makes use of Linux core features like memory management and multithreading, which is intrinsic in the java language.

- it provides the base for the application framework.

⇒ The Dalvik VM enables every android application to run in its own process, which is its own instance of the Dalvik virtual machine.

* Application framework:

⇒ It provides an environment through which we develop android applications.

⇒ It provides framework layer many higher-level services to applications in the form of java classes.

⇒ The android framework includes the following key:-

- Activity manager
- Content providers
- Resource manager
- Notifications manager
- View system

→ This runs within the android runtime with the help of services provided by all the layers.

* Applications:-

- Top layer of android architecture.

⇒ All the android application at the top layer will be grid.

Application to be installed on this layer only. The pre-installed applications downloaded from the play store like chat apps will be installed on this layer.

e.g. Contact Books, Browsers, Games

Q. Describe history of android?

2003: ^{→ october} ⇒ Android Incorporation (California)
Andy Rubin and his team.

2005 ⇒ Google acquired Android
company in \$50 million.

2007 ⇒ Google create OHA community

2008 ⇒ First android mobile was
launched, HTC

2010 ⇒ Google launched nexus series

Key employees:

- Andy Rubin
- Rich Miner
- Chris White
- Nick Sears

⇒ Android is the nickname of
Andy Rubin given by coworkers
because of his love to robots.

Q. What is OHA?

⇒ Stands for Open-handset
alliance.

⇒ It is a Consortium of 84 companies
such as Google, Samsung, Akm, KDDI,
Nokia, ~~the~~ Teleica, Ebay, Intel etc.

⇒ It was established on 5th
November, 2007 led by Google.

⇒ It is committed to advance open
standards, provide services and deploy
handsets using the android platform.

Q. What are features of Android

⇒ It is an open-source.

⇒ Anyone can customize the android
platform.

⇒ There are a lot of mobile
applications that can be chosen by
the consumer.

⇒ It provides many interesting
features like weather details, opening
screen, Live RSS feed etc.

Q. What are versions of android?

Version	Code name	API Level
1.5	Cupcake	3
1.6	Donut	4
2.1	Eclair	7
2.2	Froyo	8
2.3	Gingerbread	9 & 10
2.3.1 & 2.3.3	Honeycomb	12 & 13
4.0	Ice Cream Sandwich	15
4.1, 4.2 & 4.3	Jelly Bean	16, 17, & 18
4.4	KitKat	19
5.0	Lollipop	21
6.0	Marshmallow	23
7.0	Nougat	24 - 25
8.0	Oreo	26 - 27
9.0	Pie	
10.0	Android 10	
11.0	Android 11	
12	Android 12	
13	Android 13	
14	Android 14 (beta)	

Q. What is the current version?

⇒ Android 13.0 is the latest version of android.

⇒ it was released on August 15, 2022. it arrived first on Google pixel phones, but was quickly rolled out to Samsung Galaxy devices as well.

Q. What are categories of android applications?

⇒ There are many android applications in the market. The top categories are:

- Entertainment
- Tools
- Communication
- Productivity
- Personalization
- Music & Audio
- Social
- Media & Video
- Travel & Local

Q. What is Android Studio?

⇒ Android Studio is the official integrated development environment for android app development.

⇒ Android Studio provides more features that enhance productivity while building android apps.

⇒ Android Studio was announced on 16th May 2013 at the Google I/O Conference as an official IDE for Android app development.

⇒ The first stable version was released in December 2014. Starts from version 1.0.

⇒ Since 7th May 2019, Kotlin is Google's preferred language for android applications.

Features of Android Studio:

⇒ it has flexible Gradle-based build system.

⇒ it has fast & feature rich emulator for app testing.

⇒ it has consolidated environment where we can develop for all android devices.

⇒ Apply changes to the resource code of our running app without restarting the app.

⇒ it provides extensive testing tools & frameworks.

⇒ it supports C++ & NDK.

⇒ It provides build-in support for Google cloud platform. It makes it easy to integrate google cloud messaging & App engine.

Q. What is android studio project structure?

⇒ The android studio project contains one or more modules with resource files and source code files.

⇒ These include different types of modules:

- Android app modules
- Library modules
- Google App engine modules

Q. What is XML?

- ⇒ XML stands for extensible markup language.
- ⇒ Also known as tag-based language we can define own tags.
- ⇒ much like HTML used to describe data.
- ⇒ XML tags are not predefined.
- ⇒ it is scalable and simple to develop.
- ⇒ In Android, XML is used for front-end for designing our layouts.

Q. What are different XML files used in android?

⇒ In android there are several XML files are used for several purposes:

(1) Layout XML files:

- ⇒ To define actual UI.
- ⇒ holds all elements like (TextView, Buttons etc).
- ⇒ inside "res folder" (location)

(2) Manifest XML file (manifest.xml):

- ⇒ Used to define all the components of our application.
- ⇒ It includes the name of our application package, our Activities, receivers, services and permissions that our application needs.
- ⇒ It contains the details of our application:-

- How many activities we have created and their names
- which activity should open first.
- Permissions
- App themes, icons
- Services etc
- Package details.

⇒ it is located inside app > manifests folder.

(3) Strings XML file:

- To replace the hard coded strings with a single string.

(4) Styles XML (styles.xml):

(5) Drawable xml files:

(6) color xml file:

(7) Dimension xml (dimens.xml):

Q. What are native android applications?

- ⇒ Android phones will normally come with a suit of preinstalled applications including, but not limited to it:-
- ⇒ An SMS management applications
- ⇒ An email client compatible with Gmail but not limited to it.
- ⇒ A webkit based web browser
- ⇒ An instant messaging client
- ⇒ A music player & picture viewer
- ⇒ The Amazon MP3 store client
- for purchasing DRM free music.
- ⇒ All the native applications are written in java using the Android SDK and are run on Dalvik.

Q. What are Android SDK features?

- Some of the android SDK features:
- No licensing, distribution or development fees
- Wifi, hardware access
- GSM, EDGE, and 3G networks for telephony or data transfer allowing you to make or receive a calls or SMS messages or to send or receive data across mobile networks
- Comprehensive APIs for location-based services such as GPS.
- Full multimedia hardware control including playback and recording using the camera and microphone.
- IPC message passing
- Shared data stores
- An integrated open source webkit based browser.
- P2P support using Google Talk.
- media libraries for playing or recording a variety of audio / video or still images.

Q. What does Android run on?

⇒ The first android mobile handset, the T-Mobile G1, was released in the US in October 2008 and in the UK in November 2008.

⇒ The OpenHandsets Alliance has further committed to deploying additional handsets and services that support android in 2009 early.

⇒ Rather than a mobile OS created for a single hardware implementation, android is designed to support a very large variety of hardware platform forms, from touch-screen phones to devices with no screen at all.

⇒ Android OS is a Linux-based mobile OS that primarily runs on smartphones & tablets.

Q. What will drive for android adoption?

⇒ Several factors can influence the adoption of android:-

- User-friendly interface
- App ecosystem
- Customization
- Integration with Google service
- Affordability
- Developer friendly
- Open source nature
- Variety of devices
- Developer friendly
- Interconnectivity
- Global reach

Q. What C/C++ libraries are included in android stack?

- OpenGL - 3D graphics
- FreeType - for bitmap & vector
- SDL - for 2D graphic engine
- libc - for linux based embedded dev

SQLite: database engine to store application data.

SSL

- Support for using the secure sockets layer cryptographic protocol for secure internet communications.

Q. What are android advanced libraries?

- o android.location
- o android.media
- o android.opengl
- o android.hardware
- o android.bluetooth
- o android.net.wifi
- o android.telephony

Q. What is difference between Android virtual machine & Delvik virtual machine?

DVM

⇒ It is registered based which is designed to run on low memory.

⇒ it uses its own byte code and runs the ".dex" file.

⇒ Supports only on the android OS.

⇒ There is a constant pool for every application.

⇒ Here the executable is APK.

JVM

⇒ It is stack based.

⇒ it uses java byte code and runs ".class" file.

⇒ Supports multiple OS.

⇒ It has a constant pool for every class.

⇒ Here the executable is JAR.

Q. What are different types of android development tools?

⇒ The Android SDK includes several tools that help to test, create & debug projects.

⇒ Some are:

- The Android emulator
- DDMS (Dalvik debug monitoring services)
- Android asset packaging Tool (AAPT)
- Android Debug bridge (ADB)
- SQLite3
- Traceview
- Mksdcard
- dx
- activitycreator

Q. What is DDMS?

⇒ Stands for Dalvik debug monitoring services.

⇒ The DDMS is a debugging tool used in the Android platform.

⇒ The DDMS is downloaded as part of the Android SDK.

⇒ Some of the services provided by the DDMS are:

- Port-forwarding
- On-device screen capture
- On-device thread
- Heap-monitoring
- Radio-state information

Q. What is AAPT?

⇒ Stands for Android Asset packaging Tool.

⇒ It is a build tool that Android Studio and Android Gradle plugin use to compile and package your apps.

⇒ It is included in the tools directory of the SDK.

⇒ This tool allows:

- To view, create, and update zip-compatible archives (zip, jar, apk).
- It can also compile resources into binary assets.

⇒ It is the base builder for Android App.

Q. What is ADB?

⇒ Stands for "android debug bridge"
⇒ it is a versatile command-line tool that allows to communicate with a device.

⇒ it is a command line tool that is used:

- To debug android phone or tablet
- Send a large number of commands to control behavior on the device
- Allowing for the installation of apps
- And the logging of processes.

⇒ To achieve all that, the tool is composed of three distinct parts:

- Client
- Daemon
- Server

⇒ The Android debug bridge is included as part of Android Studio, Google's IDE for Android app development.

Q. What Android design philosophy demands?

⇒ The Android design philosophy demands that application be:

- Fast
- Responsive
- Simple
- Seamless

Q. What is AVD?

⇒ An AVD is android virtual device. it represents an Android emulator, which emulates a particular configuration of actual android device.

Q. What are different types of Android Applications?

⇒ Most of the applications created in Android fall into the following categories.

(1) Foreground Applications:

⇒ An application that's only used when it is in the foreground and is effectively suspended when it is not visible.

⇒ When creating foreground application Activity life cycle need to be considered.

⇒ Games and mashups are common examples.

(2) Background Applications:-

⇒ These applications run silently in the background with very little i/p.

⇒ They often listen for messages or actions caused by hardware, system, or other applications, rather than rely on user interaction.

⇒ Examples we include: call screening applications or SMS auto responders.

(3) Intermittent Activity:-

⇒ Expects some interactivity but does most of its work in the background.

⇒ Often these applications will be set-up and then run silently, notifying users when appropriate.

⇒ A common example would be media players app.